

Bounded Adaptive Topology-Perturbation Reweighting for Relay-Failure UAV Coordination

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Abstract

Relay-node failures in heterogeneous unmanned aerial vehicle (UAV) teams can reconfigure legal communication paths and task-support relations without necessarily causing complete information loss. We study this controlled coordination problem with bounded adaptive topology-perturbation reweighting Single-Graph multi-agent proximal policy optimization (DRTP-SG-MAPPO; DRTP). DRTP preserves the policy backbone, PPO objective, reward, execution-time information boundary, and a 50% nominal-condition anchor, and adaptively reweights six predefined failure/topology groups. Its primary comparator, Uniform Topology Randomization Single-Graph MAPPO (UTR-SG-MAPPO; UTR), has the same 116,728-parameter backbone, perturbation support, training budget, and paired evaluation protocol.

In a prospective formal cohort of five paired training seeds trained for 10M environment steps, DRTP exceeded UTR on canonical relay failure, the mean across ten frozen perturbation conditions, and the minimum over those conditions. Mean paired differences were +52.13, +55.00, and +63.01 mission-score units, respectively, and all five seeds were positive on each robustness endpoint. DRTP reduced mean timeout from 0.874 to 0.694, while collision increased from 0.005 to 0.008. In a separately completed independent three-method cohort, the DRTP-UTR direction differed and one DRTP seed met the frozen catastrophic-seed rule. Cross-tape and post hoc training-excluded-condition evaluations retained the cohort-specific directions.

DRTP therefore provides formal-cohort evidence that bounded adaptive reweighting can improve relay-failure task performance relative to matched uniform weighting. The independent outcome defines the present boundary of this evidence: the benefit has not yet replicated across separate training cohorts.

Keywords: multi-agent reinforcement learning; unmanned aerial vehicles; relay-node failure; graph coordination; adaptive training distribution; robustness

2. Problem formulation

2.1 Heterogeneous relay-failure coordination under a legal information boundary

We study a three-UAV cooperative interception task with fixed heterogeneous roles: a Scout (agent 0), a Relay (agent 1), and an Attacker (agent 2). The Scout acquires target information, the Relay can provide multi-hop communication support, and the Attacker uses legally available target information to form an attack window. The task is therefore not a collection of independent flight controllers: its feasible coordination depends jointly on motion, sensing, communication, and task-support relations.

At time t , these relations are represented by a graph $G_t=(V,E_t,X_t,Z_t)$, where V includes the three UAVs and the target, E_t contains legal sensing, communication, and task-support relations, and X_t and Z_t are node and edge features. We use a receiver-row adjacency convention: $A_t[i,j]=1$ if receiver i may legally use a relation sent by j . A sensing edge is determined only by the frozen sensing model. A communication edge must satisfy the physical range, node-state, and communication rules. A task-support edge records legally delivered information and active support; it is not an additional route for simulator ground truth.

The decentralized actor observes its own motion state and only legally available task information. In particular, a UAV that neither detects the target nor receives a fresh legal cache cannot reconstruct the true target state through graph messages. Failure labels, shortest paths, future edges, and simulator ground truth are excluded from actor input. A centralized critic is used only during training under the CTDE convention [14]; it does not enlarge the actor's execution-time information. This boundary makes the problem one of adapting coordination to legal relation changes, rather than exploiting a hidden failure indicator.

2.2 Relay-node failure as topology/path reconfiguration

A relay-node failure disables the Relay's sensing, transmission, and reception capabilities during a frozen interval beginning at t_f and lasting d_f steps, and removes the Relay-associated communication edges. Canonical F0 uses $t_f=44$ and $d_f=80$. Before the event, legal target information may follow the Scout–Relay–Attacker path. During the failure, a direct Scout–Attacker edge can remain legal when the physical rules permit it. The change is therefore

$$0 \rightarrow 1 \rightarrow 2 \rightarrow \text{quad} \rightarrow \text{quad} 0 \rightarrow 2,$$

not an assumed universal information blackout. The object of adaptation is the composition of legal communication paths, the source of task support, and the resulting coordination geometry.

The frozen training universe contains a nominal condition N, F0, and five two-member groups: early onset $TE=\{(28,80),(36,80)\}$, late onset $TL=\{(52,80),(60,80)\}$, short duration $DS=\{(44,40),(44,60)\}$, long duration $DL=\{(44,100),(44,120)\}$, and compound perturbations $CP=\{(28,120),(60,120)\}$. The tuple gives onset and duration in environment steps. These ten onset–duration members are in the training sampler support. Consequently, the cross-perturbation endpoints used in the formal study are not strict out-of-distribution (OOD) metrics. A separate post hoc evaluation on six excluded onset–duration tuples is reported only as additional unseen-member evidence.

2.3 Estimands, safety outcomes, and technical validity

For condition c , let J_c be the mean episode mission score. An episode score is the sum of stepwise rewards across the three UAVs. It includes team components for target approach, tracking, attack-window availability, legal communication connectivity, and information freshness; role components for Scout detection, Relay connectivity, and Attacker attack-window state; and penalties for energy use, collision, and constraint violations. Completion after a sustained attack window receives a terminal reward. Thus, a higher J_c indicates a more complete approach–information-support–attack-window–completion trajectory under the frozen reward contract; it is not a physical measure of distance, time, or communication throughput and cannot replace terminal safety outcomes.

The principal mission endpoints are

$$J_{\text{nominal}}, \text{quad} J_{F0}, \text{quad} J_{\text{pert,mean}} = \frac{1}{10} \sum_{c \in \text{mathcal{C}}_{\text{pert}}} J_c, \text{quad} J_{\text{pert,worst}} = \min_{c \in \text{mathcal{C}}_{\text{pert}}} J_c,$$

where $\text{mathcal{C}}_{\text{pert}}$ is the ten-member frozen perturbation family. We also report failure degradation $D_c = J_{\text{nominal}} - J_c$ and the paired training-seed difference $\Delta J_c \{D-U\} = J_c \{DTP\} - J_c \{UTR\}$. Collision rate, timeout rate, constraint-violation rate, episode length, and pre-onset collision rate are reported alongside mission score.

Failure-trigger validity is evaluated within the risk set R_c of episodes still alive at the scheduled failure onset:

$$V_{\text{trigger},c} = \frac{\#\{\text{episodes in } R_c \text{ with a correctly triggered failure}\}}{|R_c|}.$$

Episodes that terminate through a pre-onset collision remain in unconditional mission-score and safety denominators. They are excluded only from the conditional trigger-validity denominator because no failure can be injected after a legitimate termination. Training seed, rather than evaluation episode, is the independent unit for method comparison.

Relay-node failure as legal topology/path reconfiguration

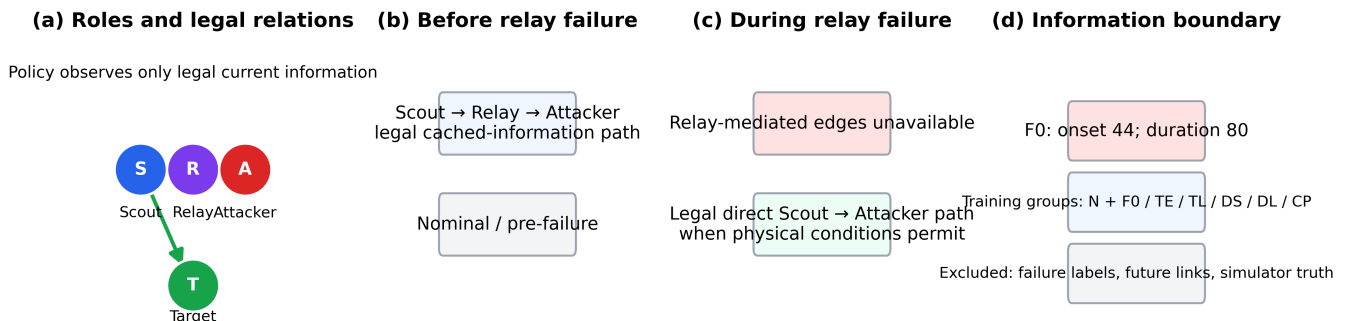


Figure 1. Relay-node failure changes the legal information and task-support path from Scout–Relay–Attacker to a direct Scout–Attacker path when that direct relation remains physically legal. This is a task-topology illustration, not a performance result.

1. Introduction

Heterogeneous UAV teams combine sensing, communication, and task-execution roles. Their cooperative capability depends not only on each vehicle's kinematics but also on whether task-relevant information can traverse legal sensing

and communication relations. Graph-based multi-agent reinforcement learning (MARL) provides representations for structured interaction [2,3,17–21], and graph neural policies have been applied to multi-UAV search, tracking, and swarm control [6,7]. A relay-node failure, however, changes the relation structure itself. When a failed relay removes a multi-hop route while a direct Scout–Attacker route remains legal, the problem is not complete information loss. It is topology/path and task-support reconfiguration under a strict decentralized information boundary.

This setting exposes a training-distribution question that is separate from graph encoding. Robust MARL studies partner, model, and adversarial variation [4], while distributionally robust reinforcement learning studies explicitly specified distributional ambiguity [5]. Group-robust learning, active domain randomization, curriculum learning, level replay, and unsupervised environment design instead study how training exposure can be allocated across groups, parameters, or generated tasks [11–13,22–26]. These approaches motivate adaptive exposure but also illustrate that changing a training distribution can change the optimization problem. They do not make a bounded heuristic sampler a minimax solver or provide a general distributionally robust guarantee for the frozen relay-failure setting.

We therefore study a deliberately narrow comparison: with the Single-Graph MAPPO backbone, PPO objective, reward, topology-perturbation support, nominal-condition exposure, budget, and paired evaluation protocol held fixed, does bounded adaptive reweighting of six predefined relay-failure groups improve mission performance relative to uniform reweighting? We ask both whether gains appear on canonical F0 and across the frozen perturbation family, and whether their direction persists in a separately completed training cohort. The second question is part of the empirical result, not a post hoc exception.

To answer this question, we introduce bounded adaptive topology-perturbation reweighting Single-Graph MAPPO (DRTP-SG-MAPPO; DRTP). DRTP retains a 50% nominal-condition anchor and redistributes the remaining failure mass across six groups using return gaps relative to nominal competence, smoothing, and bounded-simplex projection. Its primary comparator, Uniform Topology Randomization Single-Graph MAPPO (UTR-SG-MAPPO; UTR), has the same 116,728-parameter actor–critic, PPO settings, environment, reward, perturbation support, budget, and execution-time information boundary. The comparison identifies an empirical difference between bounded adaptive and uniform reweighting in this task; it does not establish necessity relative to every fixed non-uniform distribution.

Our contributions are threefold.

1. We formulate a controlled three-UAV relay-failure coordination task in which a failure can remove Relay-mediated sensing and communication relations while preserving a legal direct Scout–Attacker path. The task makes topology/path and task-support reconfiguration observable without injecting failure labels or global topology truth into decentralized actors.
2. We define DRTP as a bounded training-time reweighting controller over six frozen failure/topology groups. It leaves the actor–critic architecture, PPO objective, reward, and execution-time information unchanged, concentrating the method contrast on the training distribution.
3. We report a reliability-aware evaluation. A prospective, parameter- and exposure-matched five-seed 10M formal cohort favored DRTP on F0, J_pert,mean, and J_pert,worst, with a timeout–collision trade-off. A separately completed three-method cohort reversed the DRTP–UTR direction on both evaluation tapes. We report both strata rather than claim stable superiority or strict OOD generalization.

3. Related work and positioning

3.1 Graph-structured multi-agent learning

Communication-learning methods include continuous communication, differentiable inter-agent learning, targeted message routing, and attentional communication [17–20]. Attention can also be used in a centralized critic to select information relevant to each decentralized agent [21]. Topology-aware policy-gradient work and UAV graph-MARL studies further show that agent relations can be represented explicitly in cooperation and swarm-control problems [3,6,7]. These studies motivate the fixed Single-Graph encoder and legal information representation in our task. DRTP does not claim a new communication architecture; it holds the actor–critic fixed and studies how topology perturbations are distributed during training.

3.2 Robust learning and adaptive training distributions

Robust MARL includes minimax formulations for partner or environmental variation [4], and distributionally robust Q-learning formalizes ambiguity in a transition or environment distribution [5]. Group-robust reweighting [11], active domain randomization [12], curricula [13,25], prioritized level replay [22], and unsupervised environment design [23,24] demonstrate several ways to alter training exposure. Robust on-policy sampling likewise distinguishes the data distribution from the trajectory-collection schedule [26]. DRTP shares only the empirical premise that uniform exposure can underrepresent conditions that are difficult for a current policy. It is neither a minimax solver nor a certified group-DRO method, and it makes no worst-case guarantee. Its distinction is a reproducible, bounded six-group sampler evaluated against matched uniform sampling in one fixed relay-failure contract.

3.3 Communication-limited UAV cooperation and fault-related topology changes

MARL has been applied to UAV swarm communication under jamming [8,27,28], multi-hop relay communication [9], network recovery after UAV failure [29], and cooperative trajectory or resource allocation [30]. Communication-network and consensus studies also describe how link or node failures can require path or topology reconfiguration [15,16]. These studies establish application relevance, but their action spaces, communication objectives, reward definitions, learning algorithms, and information boundaries differ from the present relay-failure coordination task. They are used for problem positioning, not converted into nominally fair performance baselines.

3.4 Comparator boundary

The paper's primary causal comparison is UTR versus DRTP under a common SG-MAPPO backbone. MAPPO-NoGraph is reported as an external performance reference because it shares the task, seeds, budget, and evaluation tape but differs in graph-message input and parameter count. The independent cohort additionally includes a fixed non-uniform sampler, yet it has a separate cohort and evaluation tape and does not reproduce the formal DRTP direction. Neither comparison can be backfilled as evidence that online adaptation is necessary relative to every static sampler. This evidence hierarchy separates the controlled within-contract claim from external performance context and from the reliability limitation.

4. Method

4.1 Shared single-graph policy backbone and execution-time information boundary

We consider a frozen three-UAV team comprising a Scout (agent 0), a Relay (agent 1), and an Attacker (agent 2), together with one target. The environment uses lightweight three-degree-of-freedom kinematics with a 1-s control interval, a 260-step horizon, a 50-km horizontal operating radius, and an altitude range of 1–9 km. Each UAV selects one of 27 discrete commands formed by the Cartesian product of turning, climbing, and acceleration/deceleration choices. Role-specific speed, acceleration, turn-rate, climb-rate, sensing, communication, and attack-geometry limits are fixed by the environment contract.

At time t , the legal communication–task graph is

$$G_t = (V, E_t, X_t, Z_t),$$

where V contains the three UAVs and the target, E_t contains currently legal relations, and X_t and Z_t are node and edge features. The adjacency convention is receiver-row: $A_t[i,j]=1$ means that receiver i may legally use the relation sent by j . Sensing edges arise only from the frozen sensing model. Communication edges must satisfy the physical communication range, node state, and communication implementation. Task-support edges represent legally delivered information and active task support; they are not an additional channel for hidden simulator state.

The decentralized actor receives its local observation and the legal graph representation only. Its local feature vector contains self motion state, legally available relative target information, detection state, local attack-window state, energy, role, incoming connectivity, message freshness, and target-cache confidence. In particular, an agent that neither detects the target nor receives a fresh legal cache cannot recover the true target state through the graph encoder. The actor never receives a failure label, a shortest path, future link states, or simulator ground truth. A centralized critic is used during training under centralized training with decentralized execution (CTDE) [14], but it adds no execution-time information to the actor.

UTR and DRTP share one Single-Graph MAPPO actor–critic [1]. Node features are linearly encoded, processed by two shared graph-attention layers [2], fused with the local observation representation, and mapped to logits for the 27 actions:

$$a_{i,t} \sim \pi_{\theta}(a_{i,t} \mid o_{i,t}, G_{i,t}).$$

The critic estimates $V_{\phi}(s_t)$ during training. Both arms use clipped PPO [10] with generalized advantage estimation: learning rate 3×10^{-4} , discount factor 0.99, GAE coefficient 0.95, clipping coefficient 0.2, entropy coefficient 0.01, value-loss coefficient 0.5, maximum gradient norm 0.5, and four PPO epochs per batch. Each actor–critic has 116,728 trainable parameters. Thus, the primary comparison changes the training distribution rather than network capacity, the PPO objective, reward, environment, or actor information.

4.2 Frozen relay-failure groups and uniform topology randomization

A relay-node failure disables the Relay’s sensing, transmission, and reception capabilities and removes its associated communication edges from onset t_f for duration d_f . The canonical failure condition is $F_0 = (t_f=44, d_f=80)$. The event may remove the Scout–Relay–Attacker path while leaving a physical Scout–Attacker direct path legal. It therefore creates a change in legal path composition, task-support source, and coordination geometry rather than an assumed universal information blackout.

Training conditions are partitioned into one nominal group and six frozen failure/topology groups,

$$\mathcal{G} = \{N, F_0, TE, TL, DS, DL, CP\},$$

with failure subset $\mathcal{F} = \{F_0, TE, TL, DS, DL, CP\}$. The two members of each non-singleton group are fixed as

$$\begin{aligned} TE &= \{(28, 80), (36, 80)\}, \quad \& \quad TL = \{(52, 80), (60, 80)\}, \quad \& \quad DS = \{(44, 40), (44, 60)\}, \quad \& \\ DL &= \{(44, 100), (44, 120)\}, \quad \& \quad CP = \{(28, 120), (60, 120)\}. \end{aligned}$$

Here each tuple denotes onset and duration in environment steps. The nominal condition contains no injected relay failure. All failure groups have the same within-group member distribution in both arms.

Uniform Topology Randomization Single-Graph MAPPO (UTR-SG-MAPPO; UTR) fixes the nominal sampling mass at

$$p_N = 0.50.$$

It assigns a conditional uniform distribution over the six failure groups,

$$q_k^{\text{UTR}} = \frac{1}{6}, \quad \text{and} \quad p_k^{\text{UTR}} = (1 - p_N) q_k^{\text{UTR}} = \frac{1}{12}, \quad k \in \mathcal{F}.$$

UTR consequently has the same nominal anchor, failure-support universe, and within-group scenario members as DRTP. It is the parameter- and exposure-matched primary control for estimating the effect of bounded adaptive reweighting relative to uniform reweighting.

4.3 Bounded adaptive topology-perturbation reweighting

DRTP-SG-MAPPO (DRTP) is a training-time sampler controller. It preserves the total failure exposure $1 - p_N = 0.50$ and reallocates that exposure only among the six predefined groups. Its feasible conditional weights lie in the bounded simplex

$$\mathcal{Q} = \left\{ q \in \Delta^6 : 0.05 \leq q_k \leq 0.35 \right\}.$$

The controller does not alter the PPO loss, policy parameterization, critic parameterization, or execution-time inputs. It is therefore not formulated as a general min–max or distributionally robust optimization procedure, and this study makes no such guarantee.

At adaptation boundary u , let $\widehat{J}_{k,u}$ be the mean completed-episode return collected for group k since the previous boundary. For an observed group, DRTP updates its exponential moving average (EMA) as

$$\bar{J}_{k,u} = (1 - \kappa) \bar{J}_{k,u-1} + \kappa \widehat{J}_{k,u};$$

the EMA is held unchanged when that group was not observed in the window. The nominal group uses the same rule. A normalized, clipped performance gap relative to nominal competence is then

$$d_{k,u} = \text{clip} \left(\frac{\bar{J}_{N,u} - \bar{J}_{k,u}}{\max(|\bar{J}_{N,u}|, \epsilon)}, 0, d_{\max} \right),$$

followed by centering across failure groups,

$$\tilde{d}_{k,u} = d_{k,u} - \frac{1}{6} \sum_{j \in \mathcal{F}} d_{j,u}.$$

The multiplicative candidate distribution is

$$q_{k,u+1} = \frac{q_{k,u} \exp(\eta \tilde{d}_{k,u})}{\sum_{j \in \mathcal{F}} q_{j,u} \exp(\eta \tilde{d}_{j,u})}.$$

DRTP smooths this candidate and projects it onto $\mathcal{Q}$:

$$q_{u+1} = \Pi_{\mathcal{Q}} \left[(1 - \beta) q_u + \beta \tilde{q}_{u+1} \right].$$

$\Pi_{\mathcal{Q}}$ is an Euclidean projection onto the bounded simplex, not independent clipping followed by arbitrary renormalization. For a vector x , the implementation finds λ such that

$$q_k = \min\{0.35, \max\{0.05, x_k - \lambda\}\}, \quad \sum_{k \in \mathcal{F}} q_k = 1.$$

The implementation uses a deterministic bounded-simplex projection and verifies the total mass and component bounds within numerical tolerance. The projection is confined to the six-dimensional sampling state; implementation tolerances and reproducibility checks are provided in the Supplementary Information.

The frozen controller initializes $q_k=1/6$, uses 128 uniformly sampled warm-up updates, and adapts every 32 updates thereafter. Its constants are $\kappa=0.20$, $\eta=1.00$, $\beta=0.50$, $d_{\max}=2.00$, and $\epsilon=10^{-8}$. The fixed 50% nominal mass is a competence anchor that prevents the sampler from transferring all training exposure to failure conditions. The nominal EMA is a separate reference used to quantify each failure group's performance gap. Neither is an auxiliary loss or an actor/critic input.

4.4 Training and deployment boundary

At each training reset, the sampler first selects the nominal condition with probability 0.5. Otherwise it selects a failure group according to q , then samples uniformly from that group's two frozen onset–duration members where applicable. UTR keeps q fixed, whereas DRTP updates q only at the frozen adaptation boundaries described above.

Failure group labels, onset and duration values, EMA states, performance gaps, and sampler weights exist only in the training controller and its logs. They do not enter the actor or critic as new observations. The sampler is not instantiated during evaluation or deployment. DRTP consequently introduces neither trainable parameters nor an inference-time module; its additional work is restricted to per-group return accumulation, EMA updates, and a six-dimensional bounded-simplex projection. In the absence of matched hardware measurements, we report equal parameter count and no inference-time addition, rather than claiming a wall-clock or memory advantage.

The formal UTR–DRTP comparison trained both arms from scratch with four parallel environments and 64-step rollout fragments for 39,063 updates (10,000,128 environment steps) and retained only the common final 10M checkpoint for the primary endpoint. Milestone checkpoints were preserved for learning-curve visualization and interruption recovery only; they were not eligible for performance-based promotion, early stopping, seed exclusion, or reruns.

UTR-DRTP training-distribution contrast

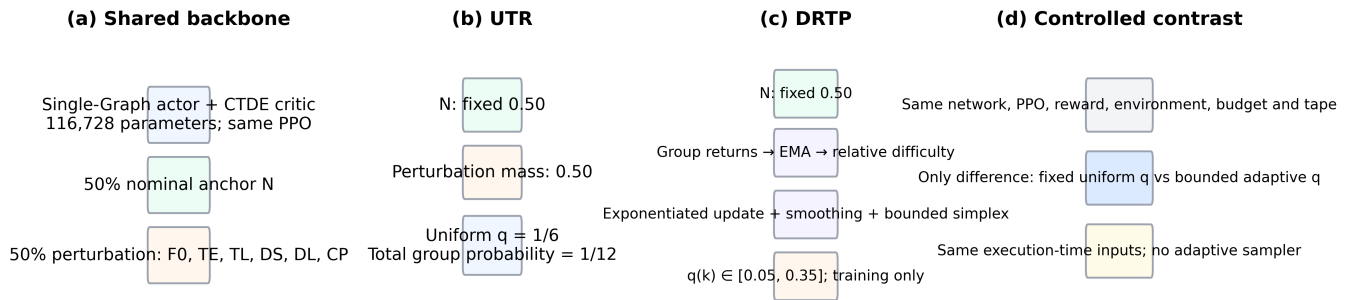


Figure 2. UTR and DRTP share the same Single-Graph MAPPO backbone and nominal-condition anchor. DRTP changes only the bounded distribution over the six frozen failure/topology groups during training.

5. Experimental protocol and results

5.1 Evidence hierarchy and formal protocol

The primary causal comparison was between DRTP and Uniform Topology Randomization Single-Graph MAPPO (UTR). The two arms used the same 116,728-parameter actor–critic backbone, PPO hyperparameters, environment, reward, seven training groups, 50% nominal-condition anchor, training budget, and execution-time information boundary. They differed only in whether the six failure/topology groups were sampled uniformly or with bounded adaptive weights. This comparison therefore estimates the empirical effect of adaptive reweighting relative to uniform reweighting in the frozen task. It does not test whether online adaptation is necessary relative to every fixed non-uniform distribution, or whether DRTP ranks above every external algorithm.

The prospective formal cohort comprised paired training seeds 2301–2305. Each UTR and DRTP trajectory was trained from scratch to a single final checkpoint at 10,000,128 environment steps; no early stopping, checkpoint promotion, seed replacement, or performance-triggered continuation was used. The paired evaluation tape contained 100 fixed base episode identifiers (490000–490099), reused across nominal, canonical F0, four onset, four duration, and two compound conditions. Each method–seed–condition cell contained 100 episodes, yielding 12,000 raw records. The evaluation tape was frozen before the formal outcomes were inspected.

Training seed, rather than episode, was the independent statistical unit ($n=5$ in the formal cohort). We report absolute mission scores, paired DRTP minus UTR differences, their mean, median, sample standard deviation (SD), interquartile range (IQR), median absolute deviation (MAD), direction count, and worst paired difference. $J_{\text{pert}, \text{mean}}$ is the mean mission score over ten frozen perturbation conditions, and $J_{\text{pert}, \text{worst}}$ is the minimum over those conditions. Because the ten onset–duration members were in the training sampler support, these are cross-perturbation endpoints, not strict OOD metrics. Collision, timeout, constraint violation, pre-onset termination, and risk-set trigger validity were reported alongside mission scores.

The formal cohort, historical development and held-out results, and the subsequent independent cohort are separate evidence strata. They are never pooled into an apparent $n=10$ confirmation experiment. A no-graph MAPPO result is reported only as an external performance reference: it shares the task, seeds, budget, and evaluation tape but has a different 35,771-parameter, no-message architecture and is not a causal drop-in baseline.

Table 1 makes the reliability boundary visible before the detailed results. The formal cohort is the prospective primary comparison. The independently completed cohort is a separate replication stratum, not an additional five seeds that can be combined with the formal cohort. Their opposite directions therefore delimit, rather than strengthen, the formal-cohort conclusion.

Evidence stratum	Arms and training seeds	Prespecified role	DRTP–UTR direction on robustness endpoints	Use in this paper
Formal primary cohort	UTR and DRTP, 2301–2305	Parameter-exposure-matched comparison and primary	Positive $J_{\text{pert}, \text{mean}}$, $J_{\text{pert}, \text{worst}}$ on F0, and	Main within-contract result
Independent reliability cohort	UTR, fixed non-uniform SNR, and DRTP, 2401–2405	Independently completed reliability stratum	Negative on the same three endpoints	Boundary on cross-cohort reliability
Paired cross-tape diagnosis	Both cohorts on tape490 and tape500	Zero-training diagnosis	Direction retained within each cohort	Excludes a tape swap as a sufficient explanation
MAPPO-NoGraph	External reference, formal seeds	Performance context	Not a matched causal contrast	Does not identify graph or sampler effects

5.2 Relay failure reconfigured legal paths rather than imposing universal blackout

The frozen mechanism audit showed that relay-node failure removed the two relay-associated communication edges during the specified failure window, while a legal direct Scout-to-Attacker edge could remain available. The information path used by the Attacker could therefore change from Scout–Relay–Attacker to Scout–Attacker. Paired diagnostic episodes also had lower task scores during failure than under the corresponding nominal condition. These observations establish the task as one of legal topology/path and task-support reconfiguration; they do not establish complete information loss or information restoration by DRTP.

5.3 Formal five-seed results favored DRTP on the primary task endpoints

At the common 10M final checkpoint, DRTP exceeded UTR on all four aggregate mission-score endpoints (Table 1). The mean timeout rate under failure conditions was lower for DRTP, whereas its collision rate was slightly higher; constraint violations were zero for both methods. Thus, the task-score result was accompanied by a timeout–collision trade-off rather than uniform safety dominance.

Method	Parameters	J_{nominal}	J_{F0}	$J_{\text{pert}, \text{mean}}$	$J_{\text{pert}, \text{worst}}$	Collision	Timeout	Constraint violation
UTR-SG-MAPPO	116,728	174.30	144.64	144.70	124.44	0.005	0.874	0.000
DRTP-SG-MAPPO	116,728	207.78	196.77	199.70	187.45	0.008	0.694	0.000

The paired seed analysis showed positive DRTP effects for all five formal seeds on F0, J_pert,mean, and J_pert,worst (Table 2). Mean paired gains were +52.13, +55.00, and +63.01, respectively; medians were +35.04, +33.00, and +38.40. The corresponding worst paired differences remained positive (+5.79, +17.01, and +11.71). In the nominal condition, four of five seeds favored DRTP. Seed 2302 had a nominal difference of -26.90 (ratio 0.760), yet remained positive on all three primary robustness endpoints. The formal overall nominal-retention rule was met (mean ratio 1.192 and median paired difference +26.66), but this cohort-level criterion was not a per-seed non-degradation guarantee. No formal DRTP seed met the frozen catastrophic-seed rule.

Endpoint	Mean paired difference	Median	SD	IQR	MAD	Positive seeds	Worst paired difference
J _{nominal}	+33.48	+26.66	47.58	51.76	34.63	4/5	-26.90
J _{F0}	+52.13	+35.04	44.99	70.60	29.25	5/5	+5.79
J _{pert,mean}	+55.00	+33.00	41.80	72.37	15.99	5/5	+17.01
J _{pert,worst}	+63.01	+38.40	50.74	87.61	26.69	5/5	+11.71

The formal degradation decomposition was consistent with the paired results. For UTR, the aggregate decreases from nominal performance were 29.66 on F0, 29.60 for J_pert,mean, and 49.86 for J_pert,worst; the corresponding DRTP decreases were 11.01, 8.08, and 20.33. These aggregate decompositions describe the cohort-level mission-score pattern and do not replace paired seed-level inference.

5.4 Gains extended across the frozen perturbation family, with bounded exceptions

The formal result was not confined to canonical F0. Mean paired differences were positive for each of the ten frozen onset, duration, and compound conditions. All four onset conditions and both compound conditions favored DRTP in 5/5 seeds. The short-duration D40 and D60 conditions favored DRTP in 4/5 seeds, with worst paired differences of -1.26 and -21.30, respectively. The other eight conditions had positive worst paired differences. This pattern supports a result across the frozen perturbation family, while retaining the condition-specific exceptions rather than treating duration effects as unconditional.

5.5 Safety and evaluator validity were reported separately

Risk-set analysis covered 5,500 planned failure episodes per method. For UTR, 5,492 episodes (99.855%) were alive at their scheduled onset and all triggered correctly. For DRTP, 5,464 episodes (99.345%) were alive at onset and all triggered correctly. Episodes absent from the risk set had terminated through pre-onset collisions and remained in unconditional task-score and safety denominators. DRTP's mean timeout rate decreased from 0.874 to 0.694, whereas its collision rate increased from 0.005 to 0.008. The collision increment was concentrated in seed 2304 (0.040 for DRTP versus 0.000 for UTR); its timeout rate also increased slightly (0.719 versus 0.701). Hence the results support a restricted statement of lower average timeout with a local collision cost, not that DRTP is uniformly safer.

Mission-score patterns were aligned with terminal outcome summaries: for example, F0 completion increased from 0.116 for UTR to 0.278 for DRTP and the corresponding timeout rate decreased from 0.880 to 0.714. Collision remained low in absolute terms but was generally slightly higher for DRTP across failure condition families.

5.6 Sampler telemetry confirmed changed training exposure but not a causal policy mechanism

The adaptive sampler departed from uniform weighting in every formal DRTP run. At its final update, the average weights for compound perturbations, long durations, and early-onset failures were 0.273, 0.237, and 0.203, respectively, above the uniform six-group weight of one sixth. The corresponding realized training episode counts were 26,804, 21,619, and 19,226; F0, late-onset, and short-duration groups had lower counts, while the fixed nominal anchor contributed 96,822 episodes. In the final failure evaluation, DRTP showed a higher task-support proportion (0.630 versus 0.508), a higher legal-information proportion (0.892 versus 0.802), lower cache age (16.62 versus 18.96), and fewer path switches (6.56 versus 8.06) than UTR.

These telemetry observations verify that DRTP changed realized exposure and are consistent with different path/task-support utilization. They do not identify a causal policy mechanism, and are therefore not used as an additional performance test.

5.7 External reference and independent reliability evidence delimited the result

MAPPO-NoGraph was evaluated as an external performance reference. DRTP's mean paired differences relative to it were +1.20 on J_{nominal} , +3.08 on F0, +9.63 on $J_{\text{pert, mean}}$, and +7.68 on $J_{\text{pert, worst}}$; all four endpoints were positive in 4/5 seeds. MAPPO-NoGraph nevertheless had lower collision and timeout rates (0.002 and 0.601 versus 0.008 and 0.694 for DRTP). Because its message inputs and parameter count differ, this result locates performance in the frozen task but does not isolate graph structure, capacity, or adaptive weighting effects.

The independent 2401–2405 three-method cohort used UTR, a fixed non-uniform SNR comparator, and DRTP under the same 10M endpoint rule but with a separate evaluation tape. In this cohort, UTR exceeded DRTP on every aggregate task endpoint. The mean paired DRTP-minus-UTR differences were -38.35 on J_{nominal} , -33.27 on F0, -34.07 on $J_{\text{pert, mean}}$, and -32.36 on $J_{\text{pert, worst}}$; only 1/5 or 2/5 seeds were positive, and one DRTP seed met the catastrophic rule. DRTP also had higher failure-condition collision and timeout rates than UTR (0.052 and 0.678 versus 0.009 and 0.646). SNR did not outperform UTR in this independent cohort, so the comparison did not establish either a fixed non-uniform or an online adaptive distribution as necessary.

Cross-tape evaluation separated the cohort effect from a simple evaluation-tape explanation. The formal cohort remained positive on both tape490 and tape500: the DRTP-minus-UTR means were +52.13/+55.00/+63.01 on tape490 and +60.25/+61.07/+68.31 on tape500 for F0, $J_{\text{pert, mean}}$, and $J_{\text{pert, worst}}$. The independent cohort remained negative on both tapes, with the corresponding means -33.58/-35.37/-37.10 and -33.11/-34.07/-32.36. This diagnosis localizes the reversal to the training-cohort/seed level but does not identify a specific random source or policy-basin mechanism.

Finally, a post hoc, zero-training additional evaluation used six training-excluded onset–duration tuples and a separate deterministic tape. The formal cohort retained positive median directions across all six conditions (mean paired differences from +35.36 to +70.85), whereas the independent cohort retained negative median directions (from -25.20 to -49.46). This result is additional unseen-member evidence rather than a preregistered confirmatory OOD test.

Formal paired five-seed comparison at 10M

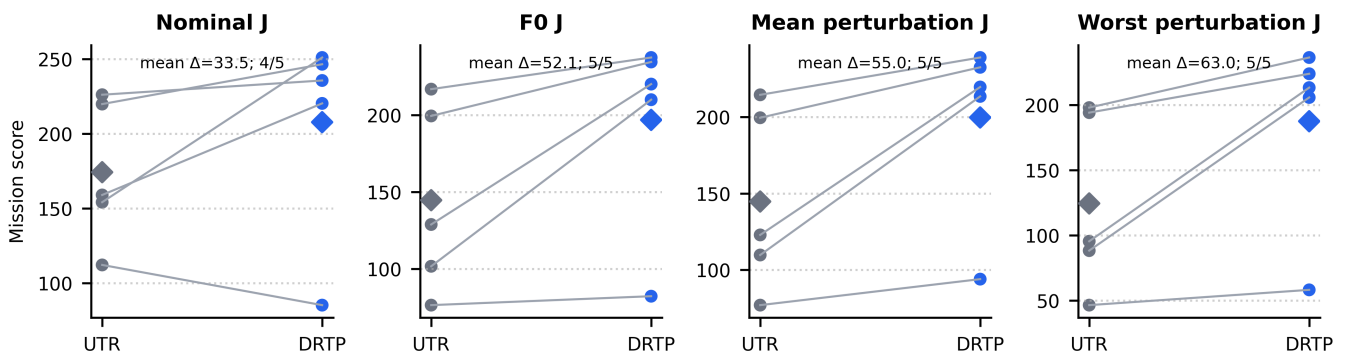


Figure 3. Formal paired five-seed performance at the common 10M checkpoint. Points show all training seeds; training seed is the independent unit.

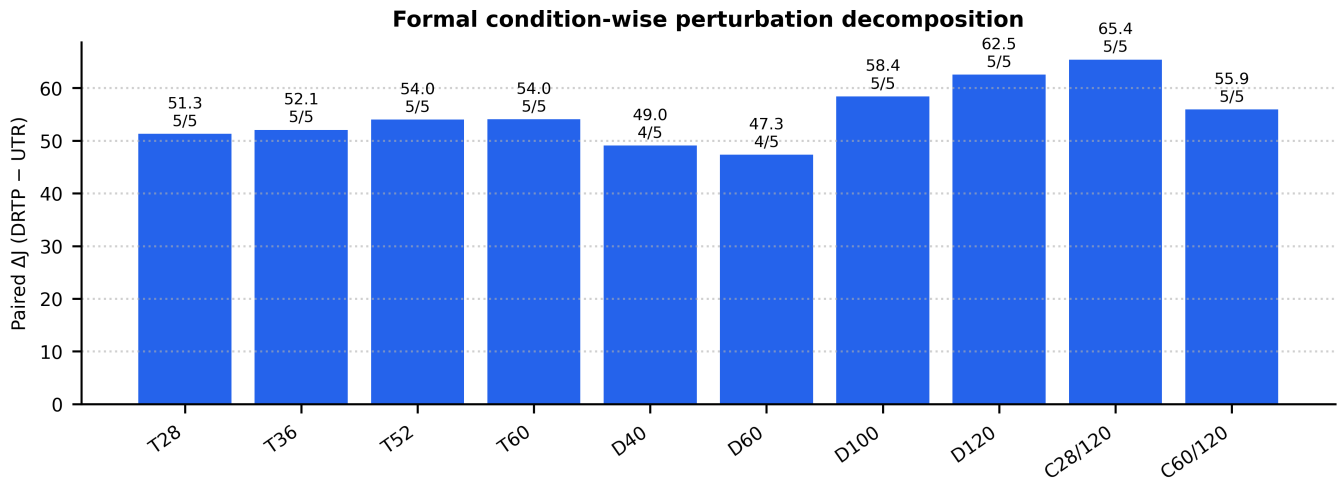


Figure 4. Formal condition-wise decomposition over the ten frozen perturbation members. These members were within training support and are not presented as strict OOD results.

Formal reliability and safety boundary

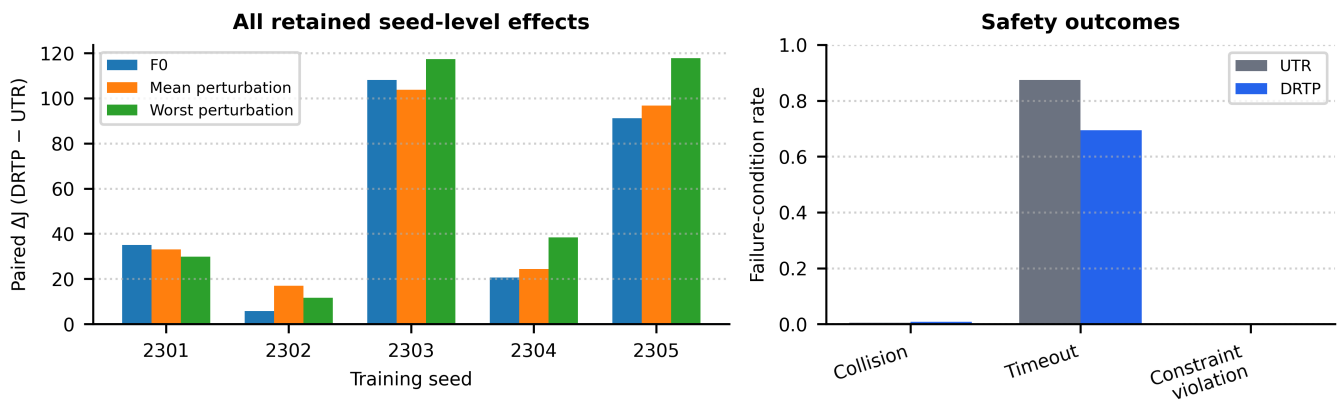


Figure 5. Formal seed-level reliability and safety audit. The panel separates mission-score outcomes from collision, timeout, and failure-trigger validity.

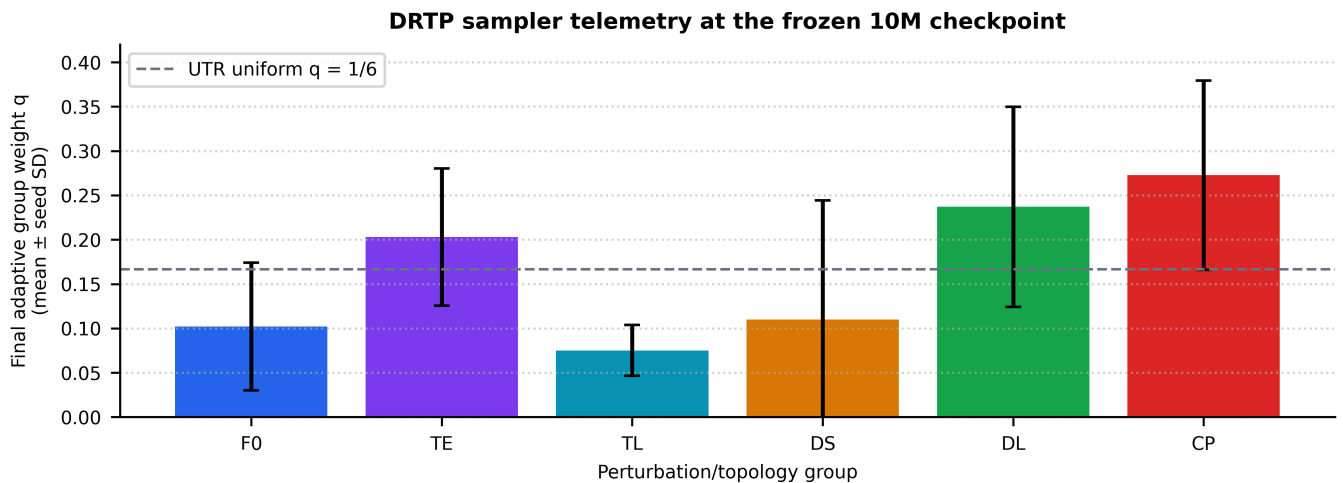


Figure 6. D RTP sampler telemetry. These records verify altered training exposure but do not establish a causal policy mechanism.

Formal terminal outcomes

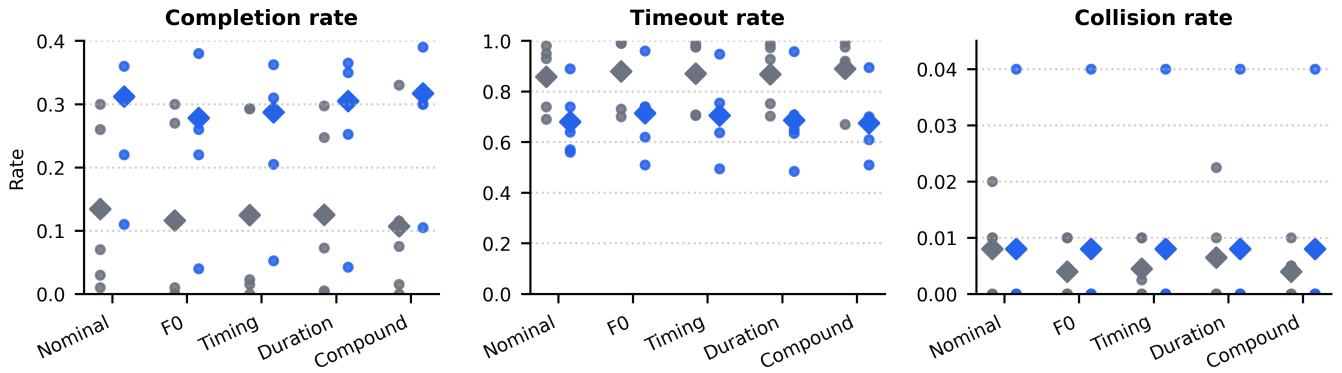


Figure 7. Formal terminal outcomes. Mission-score gains coincide with greater completion and lower timeout, while collision is reported separately as a trade-off.

6. Discussion

6.1 What the formal comparison establishes

The contribution of DRTP is deliberately narrow. With model capacity, PPO, reward, perturbation support, nominal exposure, and execution-time information fixed, the formal cohort showed that bounded adaptive reweighting of failure groups outperformed uniform reweighting on canonical F0 and the two frozen cross-perturbation endpoints. The result was not driven solely by a higher nominal score: the aggregate degradation from nominal performance was also smaller for DRTP. However, the formal nominal-retention rule was cohort-level; the nominal decline in seed 2302 shows that it must not be read as a guarantee for every initialization.

The formal experiment does not establish that online adaptation is superior to all static non-uniform distributions. A fixed non-uniform comparator was not trained concurrently in the primary cohort, and the later SNR comparator did not outperform UTR in the independent cohort. DRTP is consequently an empirical, cohort-bounded training-distribution intervention rather than a general distributionally robust optimization method.

6.2 Training-cohort sensitivity limits generalization of the gain

The independent three-method cohort reversed the direction of all four aggregate DRTP-minus-UTR task endpoints and contained a catastrophic DRTP seed. This observation does not invalidate the completed formal comparison: its five-seed protocol, final checkpoints, and positive paired results remain factual. It does limit the scope of any general claim. In particular, the formal outcome cannot be promoted to stable superiority across training cohorts, random initializations, or arbitrary training realizations.

The cross-tape diagnosis makes an evaluation-tape explanation insufficient. Both cohorts retained their respective directions when evaluated on tape490 and tape500. The additional evaluation on six training-excluded onset–duration members likewise reproduced positive formal-cohort and negative independent-cohort directions. These checks localize the unresolved issue to the training-cohort/seed level. They do not demonstrate that a specific random source, optimization basin, sampler excursion, or policy mechanism caused the reversal.

6.3 Reliability stress tests strengthen the boundary rather than add a method claim

Exploratory stabilization studies were conducted after the formal comparison to test whether the observed risk could be removed by readily available local interventions. Bounded sampler steps, a uniform anchor, fixed KL-based actor protection, training-only paired probes, candidate selection, and a counterfactual critic each produced local positive signals in some development seeds or pilots. None retained task gain, upper-tail behavior, and reduced seed dispersion in its subsequent independent cohort. The counterfactual critic produced systematic task degradation in two fresh five-seed cohorts.

These negative studies do not constitute alternative main methods, are not pooled with the formal DRTP evidence, and do not identify a failure mechanism. Their narrower value is to rule out the claim that the independent reversal is already resolved by a simple sampler, KL, probe, selector, or critic patch. We therefore retain training-cohort sensitivity as a current

applicability boundary instead of selectively reporting a locally favorable stabilization candidate. A compact evidence ledger is provided in Supplementary Table S5.

6.4 Safety, information-path interpretation, and alternative explanations

Higher mission score did not imply uniform safety improvement. In the formal cohort, DRTP reduced mean timeout but slightly increased collision, with the increment concentrated in one seed. The result should consequently be read as a task–safety trade-off. Risk-set trigger validity established that failures were correctly injected for all episodes alive at their scheduled onset; it did not remove pre-onset collisions from the task or safety denominators.

The matched UTR control also excludes one simple exposure explanation: both methods encountered the same seven training groups and shared the same nominal anchor, so the formal contrast cannot be explained by DRTP seeing a failure class unavailable to UTR. Sampler telemetry confirmed changed training exposure and was consistent with altered task-support/path utilization. It did not demonstrate that a particular weight change caused a particular policy behavior or restored unavailable information. Relay failure can leave a legal direct Scout-to-Attacker path intact; the problem is therefore topology/path reconfiguration, not universal communication blackout.

6.5 Scope and future research

The evidence is limited to a frozen three-UAV, lightweight 3DOF simulation with predefined relay-failure conditions. It contains neither larger teams, hardware-in-the-loop validation, nor flight tests. The primary causal evidence is the matched UTR–DRTP comparison; MAPPO-NoGraph is an external performance reference with different capacity and inputs. The ten primary perturbation members were seen by the training sampler, while the six excluded-member tests were post hoc and cannot be presented as confirmatory OOD generalization.

Future reliability work requires a new mechanism-focused contract rather than another result-driven stabilizer. Such a contract would need a repeated, time-leading, DRTP-specific precursor and a single intervention that targets it, followed by prospective validation in separate fresh-seed cohorts. Until then, the appropriate interpretation is that DRTP has substantial formal-cohort upside under the stated contract, with an unresolved cross-cohort reliability limitation.

7. Conclusion

This study examined heterogeneous UAV coordination when a relay-node failure reconfigures legal communication paths and task-support relations. DRTP-SG-MAPPO retained the Single-Graph MAPPO backbone, PPO objective, reward, and execution-time information boundary, while using bounded adaptive reweighting over six predefined failure/topology groups under a fixed nominal-condition anchor. Its matched UTR-SG-MAPPO control used the same parameter count, perturbation support, training budget, and evaluation protocol, isolating adaptive rather than uniform training-group weighting as the intended contrast.

In the prospective formal five-seed cohort, DRTP had positive paired effects on canonical F_0 , $J_{\text{pert,mean}}$, and $J_{\text{pert,worst}}$, with all five seeds positive on these primary endpoints. It also had a lower mean timeout rate but a slightly higher collision rate, so the formal result is a mission-score improvement with a safety trade-off rather than evidence of uniform safety improvement. A post hoc evaluation on six training-excluded onset–duration members retained the positive direction in the formal cohort. MAPPO-NoGraph provided external task context but, because of its different inputs and capacity, did not replace the matched UTR–DRTP comparison.

The independent three-method cohort produced a different DRTP–UTR direction and included a catastrophic DRTP training seed. Cross-evaluation-tape and additional unseen-member diagnostics retained these cohort-specific directions, making a simple evaluation-tape explanation insufficient. DRTP is thus positioned as a bounded training-time strategy with demonstrated formal-cohort benefit under the frozen contract and an explicit cross-cohort reliability boundary. This combination of controlled task semantics, matched evidence, and transparent reliability reporting provides a rigorous basis for future mechanism-focused studies of adaptive MARL training reliability.

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Data Availability Statement

The local anonymous reproducibility package has passed its technical staging checks. AUTHOR_INPUT_NEEDED: a live reviewer-access URL, licence, checkpoint/runtime-state access policy, and external download verification before submission.

Conflicts of Interest

AUTHOR_INPUT_NEEDED: conflict-of-interest declaration confirmed by all authors.

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